

① $q = 1 \mu C$
 $V = 10^{-6} C$ $X = 8 cm$ $E = \text{force due to the charge } +q \text{ at } +$

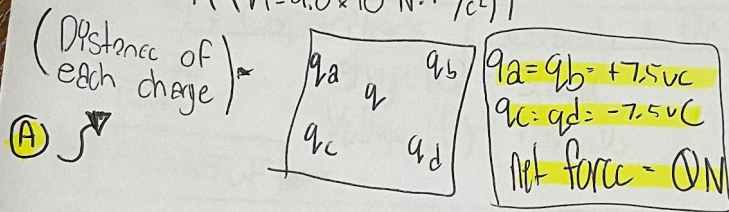
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 PHYS 135B

A. $F = k \frac{q \times 2q}{.03^2} - k \frac{q \times 2q}{.05^2}$
 $= \frac{9 \times 10^9 \times 10^{-6} \times 2 \times 10^{-6}}{.03^2} - \frac{9 \times 10^9 \times 10^{-6} \times 2 \times 10^{-6}}{.05^2}$
 $= 20 - 7.2$
 $F = \boxed{12.8 N}$
 R.

B. $E = E_{-2q} + E_{+q} + E_{+3q}$
 $= k \times 2q + k \times q + k \times 3q$
 $= \frac{-9 \times 10^9 \times 2 \times 10^{-6}}{.1^2} + \frac{9 \times 10^9 \times 3 \times 10^{-6}}{.03^2}$
 $E = \boxed{3.07 \times 10^7 N/C}$
 L.

② Given: $(q_a = q_b = +7.50 \mu C)$
 $(q_c = q_d = -7.50 \mu C)$
 $(q = 2.00 \mu C)$
 Square side length
 $10 cm = .10 m$
 Coulomb's Constant
 $(k = 9.0 \times 10^9 N \cdot m^2/C^2)$

③ $q = 2 \mu C$, $L = .11 m$ $k = 9 \times 10^9$
 distance from point to q
 $r = \frac{L}{\sqrt{2}} = \frac{.10}{\sqrt{2}} = 0.0707 m$
 $F = k \frac{q_1 q_2}{r^2} = k \frac{(2 \times 10^{-6})^2}{(.0707)^2}$
 $= 27.0 N$
 however since they are symmetrical, the force on q is $0 N$



③ $40 kV$ $qV = \frac{1}{2} mv^2$ $V = \frac{2qV}{m}$ $q = 1.6 \times 10^{-19}$
 $m = 9.11 \times 10^{-31} kg$ $V = 40000 = 4 \times 10^4$

$V = \sqrt{\frac{2qV}{m}}$ $V = \sqrt{\frac{2 (1.6 \times 10^{-19}) (4 \times 10^4)}{9.11 \times 10^{-31}}}$
 $V = \boxed{1.14 \times 10^8 m/s}$

④ A $\Delta V = E \cdot d = 7.5 \times 10^4 (.04 cm)$
 $= \boxed{3.0 \times 10^3 V}$

B $1 cm$ $33 cm$
 $= 7.5 \times 10^4 (.01)$ $= 7.5 \times 10^4 (.03)$
 $= \boxed{750 V}$ $= \boxed{2250 V}$

C $\frac{\Delta V}{d} = E$ $\Delta V = 80 mV$ $E = \frac{80 \times 10^{-3}}{9 \times 10^{-9}}$ $\boxed{8.9 \times 10^6}$

⑤ Given

$$E = 32 \text{ KEV}$$

$$D = 2.00 \text{ nm}$$

$$32 \text{ KeV} = 32 \times 10^3 \times 1.6 \times 10^{-19}$$

$$= 5.12 \times 10^{-16} \text{ J}$$

$$q = ze$$

$$= 2 \times 1.6 \times 10^{-19}$$

$$= 3.2 \times 10^{-19} \text{ C}$$

$$E = V/m \quad \frac{1.6 \times 10^4}{2} = 8 \times 10^3 \text{ V/m}$$

$$V = E/D = \frac{5.12 \times 10^{-16}}{3.2 \times 10^{-10}} = 1.6 \times 10^{-6} \text{ V}$$

$$8 \times 10^3 \text{ V/m}$$

⑥ 5.00 MeV, 79 (protons)

$$U(r) = \frac{k \cdot Z_1 q_1 \cdot Z_2 q_2}{r}$$

$$r = \frac{k \cdot Z_1 q_1 \cdot Z_2 q_2}{E}$$

$$E = 5 \text{ MeV} = 5 \times 10^6 \text{ eV}$$

$$Z_1 = 79 \text{ protons}$$

$$Z_2 = 2$$

$$k = 8.99 \times 10^9$$

$$e = 1.602 \times 10^{-19} \text{ C}$$

~~Wanted~~

$$r = \frac{(8.99 \times 10^9)(79)(2)(1.602 \times 10^{-19})^2}{(5 \times 10^6)(1.602 \times 10^{-19})} = \frac{(1.846 \times 10^{-25})}{8.01 \times 10^{-13}} \approx 2.3 \times 10^{-15} \text{ m}$$

$$= \frac{(1.846 \times 10^{-25})}{8.01 \times 10^{-13}} \approx 2.3 \times 10^{-15} \text{ m}$$

Unit #1: Capacitors, Current, & DC circuits

① $C = \frac{3.00 \mu\text{C}}{120 \text{ V}}$

charge $Q = 3.00 \mu\text{C}$

Voltage $(V) = 120 \text{ volts}$

$\rightarrow 0.025 \mu\text{F}$

② $C = 10 \mu\text{F}$ $V = 9 \times 10^3 \text{ V}$ $U = \frac{1}{2} CV^2$ $Q = CV$

① $= .5 \times 10 \times 10^{-6} \times (9 \times 10^3)^2$

$= 405 \text{ J}$

② $8 \mu\text{F} / 40 \text{ J}$

$E = \frac{1}{2} CV^2$

$V = \sqrt{\frac{2E}{C}}$

$V = \sqrt{1 \times 10^7}$

$V = \sqrt{\frac{2(40)}{8 \times 10^{-6}}}$

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$V = \sqrt{\frac{2(40)}{8 \times 10^{-6}}}$

③ ④ Capacity is $= E = \frac{1}{2} (40) V^2$

$V = 3162.3$

~~Wanted~~ 405 J $V = Q \times 10^3$ $405 = 20(4 \times 10^{-6})^2$

$405 = 1.62 \times 10^3 \text{ C}$

$C_{\text{tot}} = C_1 + C_2 + C_3 + C_4 = 4 \text{ C}$

$C = 405 / 1.62 \times 10^3$

$= 2.5 \mu\text{F}$ $Q = 2.5 \times 10^{-6} \times 4 \times 10^3$

$= 2.5 \text{ mC}$

④ In parallel is much
charge versus in series. more

better charge

④ 3.0V battery, 13Ω

⑤ flow of battery, 3Ω

$R_{LED} = 3\Omega$, $R_{res} = 1k\Omega = 1000\Omega$, $V = 3V$

⑥ $Leb = LED$

$$P = I^2 \cdot R_{LED}$$

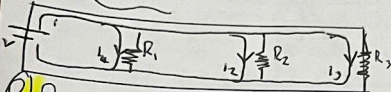
$$P = (0.00299A)^2 \cdot 3\Omega = 2.68 \times 10^{-6}W$$

$$= 8.94 \times 10^{-6}A (3\Omega) = 268 \mu W$$

$$I = V/R_{tot} = \frac{3V}{1003\Omega} = 0.00299A = 2.99mA$$

Unit #2 DC Circuits with resistors in series & parallel, R_L circuits

① Figure #3



③ $R_{tot} = 2000\Omega$, $V = 12V$

$$I = V/R_{tot} = 12V/2000 = 0.006A = 6mA$$

$$\frac{1}{R_{tot}} = \frac{1}{1000} + \frac{1}{500} + \frac{1}{R_3}$$

$$\frac{1}{R_{tot}} = \frac{3}{1000} + \frac{1}{R_3}$$

$$\frac{1}{R_3} = \left(\frac{1}{R_{tot}}\right) - \frac{3}{1000} = \frac{5}{1000}$$

$$\frac{1}{R_3} = \frac{5-3}{1000} = \frac{2}{1000} \Rightarrow R_3 = 500\Omega$$

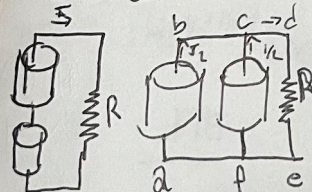
$$\frac{12V}{1000} = 0.012A \quad 1.2mA = I_1$$

$$\frac{12V}{500} = 0.024A \quad 2.4mA = I_2$$

$$\frac{12V}{5000} = 2.4mA = I_3$$

② Figure #4

DC circuit with 3 resistors



$R = .5k\Omega$

$$\textcircled{A} R = .5k\Omega = 500\Omega \quad I = V/R = 3V/500 = 0.006A$$

$$V_{series} = 1.5 + 1.5 = 3V$$

$$V_{parallel} = 3V$$

Series

Parallel

$$P_{series} = I^2 R = (0.006)^2 (500) = 18mW$$

$$\textcircled{B} R = .5\Omega \quad P = I^2 R = (0.006)^2 (500) = 18mW$$

$$I = V/R = 3V/500\Omega = 0.006A$$

$$V = 3V, R = 1\Omega$$

$$P = I^2 R = (0.006)^2 (500) = 18mW$$

$$I = V/R = 3V/500\Omega = 0.006A$$

$$P = I^2 R = (0.006)^2 (500) = 18mW$$

$$I_{parallel} = 3V/500\Omega = 0.006A$$

$$P_{parallel} = (0.006)^2 (500) = 18mW$$

$$.5/2 = .25$$

$$I = \frac{3V}{500.25} = 0.00599A$$

$$P = I^2 R = (0.00599)^2 (500) = 17.94mW$$

$$5.99mA$$

$$5.99mA$$

$$17.94mW$$

$$I = \frac{E_2 + 3E_1}{r_2 + 3r_1 + r_2} = \frac{1.53 + 3(1.53)}{.10 + (3)(.02) + .10}$$

$$I = .617 \text{ A}$$

$$P = I^2 R_L = (.617 \text{ A})^2 (10 \Omega) = 3.806 \text{ W}$$

$$P = I^2 R_L \quad I = \frac{E_2 + 3E_1}{r_2 + 3r_1 + R_L} \quad r_2 = \frac{[3(1.53) + 1.53V] - .224 [E_2 + 1.53V]}{.224}$$

$$I = 17.9 \Omega$$

Unit #3 Magnetism I

Case	V direction	B direction	F direction
(a)	right	up	in page
(b)	up	out page	left
(c)	out of page	right	down

~~Answer in chart~~
(Answer in chart)

$$4.00 \times 10^3 \text{ m s}^{-1}, 1.25 \text{ T}, 1.40 \times 10^{-16} \text{ N}$$

$$F = qvB \sin \theta$$

$$\sin \theta = F/qvB$$

$$\sin \theta = \frac{1.4 \times 10^{-16}}{(1.6 \times 10^{-19})(4 \times 10^3)(1.25)}$$

$$= \frac{1.4 \times 10^{-16}}{8 \times 10^{-16}}$$

$$= .175$$

$$\theta = \arcsin(.175)$$

$$\theta = 10.8^\circ \text{ or } 169.2^\circ$$

$$180 - 10.8$$

$$169.2$$

Soh Cah Toa
 $\sin = o/h$
 $\cos = a/h$
 $\tan = o/a$
 $\arcsin$
 $\arccos$
 $\arctan$
 $\sin = o/h$
 $\cos = a/h$
 $\tan = o/a$
 $E = W$

③ given

Oxygen - 16 protons

$$m_{255} = 2.66 \times 10^{-26} \text{ kg}$$

$$V = 5.00 \times 10^6 \text{ m/s}$$

$$B = 1.20 \text{ T}$$

$$r = 231 \text{ m}$$

① $r = \frac{mv}{qB}$

$$= \frac{(2.66 \times 10^{-26} \text{ kg})(5.00 \times 10^6 \text{ m/s})}{(1.231 \text{ m})(1.20 \text{ T})}$$

$$= 4.8 \times 10^{-19} \text{ C}$$

$$Q = \frac{4.8 \times 10^{-19} \text{ C}}{1.6 \times 10^{-19}}$$

$$= 3e$$

② $e = 1.6 \times 10^{-19} \text{ C}$

$$\frac{Q}{e} = 3e/e \quad \boxed{3/1}$$

③ $m_{18} = m_{16} \times 18/16$

$$= 18/16 \times 2.66 \times 10^{-26} \text{ kg}$$

$$= 2.99 \times 10^{-26} \text{ kg}$$

$$r' = \frac{m_{18} v}{qB}$$

$$\frac{m_{18}}{m_{16}}$$

$$(231 \text{ m}) \times \frac{2.99 \times 10^{-26}}{2.66 \times 10^{-26}}$$

$$= 26 \text{ m}$$